MEMORANDUM

To: Professor Kathryn Schulte Grahame
CC: Teaching Assistant, GE1501/1502
From: Honghan Zheng HZ
Date: October 18, 2026
Re: AutoCAD Lab #4 - Laser-Cut LCD Enclosure

This memo describes the enclosure I designed for my 16x2 LCD and I2C backpack, and how to recreate it from my drawing, Zheng_AC4.pdf. The enclosure is shaped like Clawd, the pixel mascot from Claude Code [1]. I picked it because I plan to use the screen to show my daily token usage. The enclosure is made of ten 1/8 in. laser-cut layers stacked together, and it covers the LCD on all six sides.

Design and function. All ten layers use the same 150 x 90 mm Clawd outline. I drew it on a 10 mm grid, so it only has straight lines, and the four legs work as a stand. What changes between layers is the opening in the middle (Fig. 1). The front plate has a 66.7 x 16.5 mm window. This is 1 mm bigger than the viewing area on each side, so the screen shows but the module cannot fall out the front. The eyes are cut through, which leaves two 3 mm deep holes. Behind the front plate are three bezel layers with a 71.7 x 24.7 mm pocket for the metal bezel, then five PCB layers with an 80.6 x 36.6 mm pocket for the board and the backpack, plus a small bay for the wire connectors. The back plate has a 31 x 14 mm window over the four pins, so the wires can be connected while the screen is in the box. It also has a 4 mm hole for the LED and a 5 mm hole for turning the contrast knob (Fig. 2). Four 6.4 mm holes go through every layer for M6 bolts.

Conceiving and planning the design. My first idea was a regular box, but a box is hard to fit exactly because of the finger joints and the kerf. The LCD with the backpack is about 18 mm thick, and the laser can only cut flat sheets. So I decided to stack flat layers instead. Each layer is just a 2-D shape, and the pockets hold the screen in place. I took all the LCD dimensions from the course file LCD.dwg [2] and compared them with a 1602 datasheet [3]. I left 0.3 mm of clearance on each side of every pocket. To get the number of layers, I divided the height of each part by the sheet thickness and rounded up. The bezel is about 8 mm, so it needs three layers. The board and backpack are about 13 mm together, so they need five. In AutoCAD I drew the outline with PLINE (closed with C), the pockets with RECTANG, and the holes with CIRCLE. Then I used COPY to make the other layers.

Planning for laser cutting. I put the cut lines on a red CUT layer, and the dimensions and leaders on a separate DIM layer outside the parts. The ten parts fit on two 12 x 18 in. sheets with 7 mm gaps and 10 mm margins, and each sheet is its own 1:1 layout. Before exporting, I made sure every cut line was a closed polyline, there were no double lines, and no wall was thinner than 3 mm.

Assembly and finishing. To put it together, sand the burnt edges a little and stack the layers in order, using the bolts to line them up. The front plate and bezel layers are glued with wood glue. The LCD goes in from the back, then the back plate goes on and four M6 bolts and nuts hold everything together. The bolts are 3D printed in M6, because M3 threads are too small to print well on an FDM printer [4]. Before cutting, I will measure the real sheet thickness with calipers. If it is thinner than 2.9 mm, I will add a sixth PCB layer. If the LCD feels loose, a small piece of foam tape behind the board will keep it in place.

Revisions. I made a few changes along the way. At first I thought LCD.dwg showed the back of the module, so I mirrored the coordinates. This put the pin window and the LED hole on the wrong side. I fixed it after I noticed the 16-pin header was in the top-left corner, which means the drawing is a front view. I also went from four PCB layers to five, because some backpacks are up to 15.3 mm tall. When I cut, I plan to cut the front plate and one bezel layer first and test them with the LCD. If the pocket is too tight, I will open the clearance from 0.3 mm to 0.4 mm before cutting the rest.

Thank you for reviewing my design. The stacked layers cover all six sides of the LCD, hold it without gaps, and still leave the pins and the LED easy to reach.

References

[1] Anthropic, "Claude Code," 2026. [Online]. Available: https://www.anthropic.com/claude-code. Accessed: Oct. 18, 2026.

[2] GE1501/1502 Course Staff, "LCD.dwg," AutoCAD drawing, Northeastern University, Boston, MA, USA, 2026.

[3] "SC1602A LCD module datasheet," AllDatasheet. [Online]. Available: https://www.alldatasheet.co.uk/html-pdf/89725/ETC/SC1602A/46/1/SC1602A.html. Accessed: Oct. 18, 2026.

[4] Sovol3D, "3D printing threads and screws: How to design reliable FDM fasteners." [Online]. Available: https://www.sovol3d.com/blogs/news/3d-printing-threads-and-screws-how-to-design-reliable-fdm-fasteners. Accessed: Oct. 18, 2026.

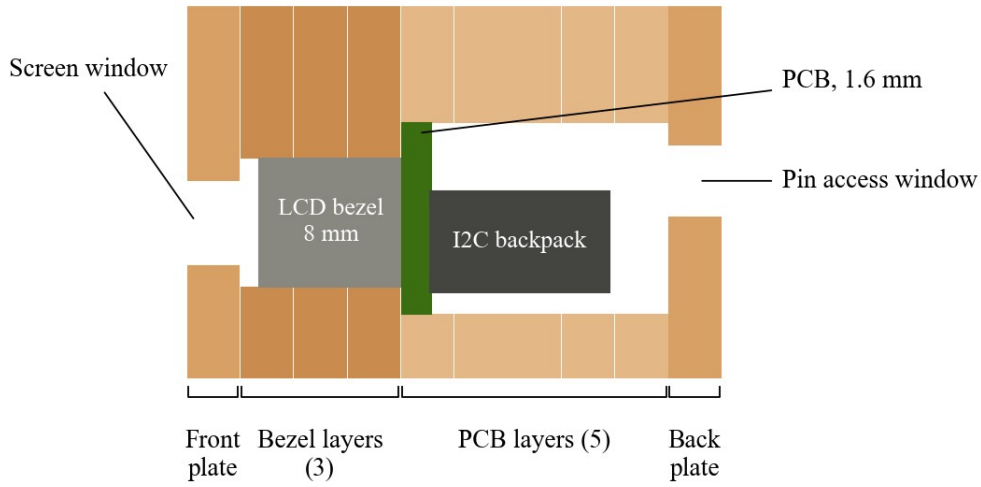


Fig. 1. Side section of the layer stack. The bezel layers locate the LCD bezel, the PCB layers hold the board and backpack, and the back plate leaves the four pins exposed.

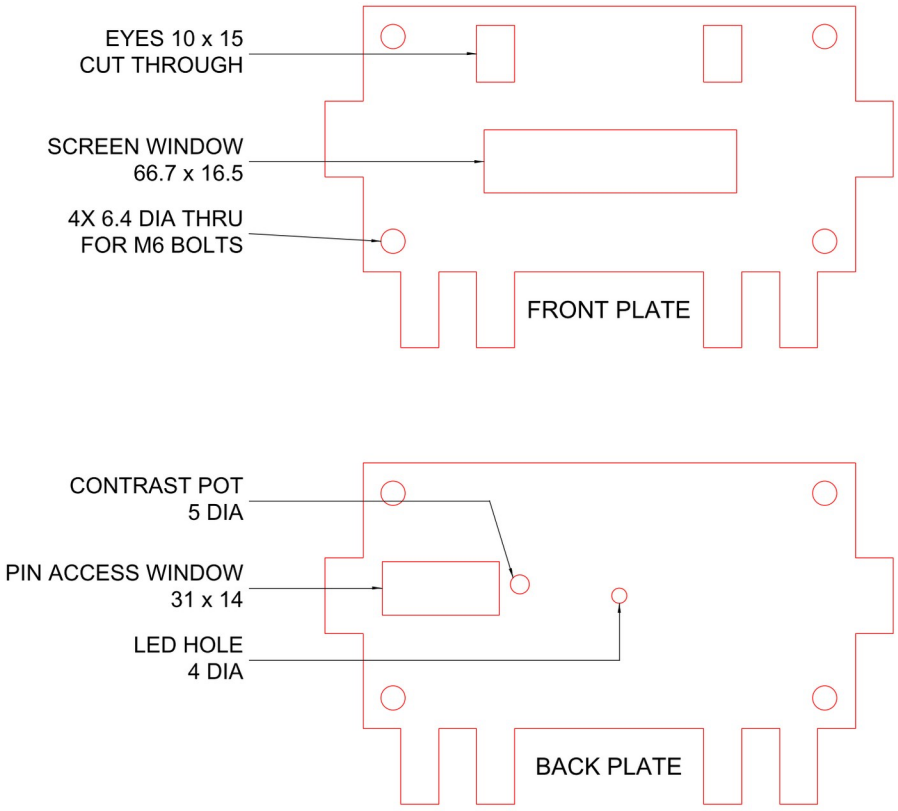


Fig. 2. Front plate (top) and back plate (bottom) from Zheng_AC4.pdf, Sheets 1 and 2.